

Daily Problem: NFA Square Problem is NP-Complete

Problem Statement

Given an NFA A , decide whether there exists a non-empty string w such that ww is in $L(A)$. Prove that this problem is NP-complete.

Solution

We prove NP-completeness by showing membership in NP and NP-hardness via reduction from Hamiltonian Cycle.

Step 1: In NP

Guess a string w and verify that ww is accepted by A using standard NFA simulation. This can be done in polynomial time.

Step 2: Reduction from Hamiltonian Cycle

Given a graph $G = (V, E)$, construct an NFA that accepts strings of the form $v_1 \dots v_n \# v_1 \dots v_n$ where the sequence forms a Hamiltonian cycle.

Step 3: Construction idea

The NFA checks vertex uniqueness, edge validity, and matching halves using nondeterminism and state tracking.

Step 4: Correctness

Such a string exists iff G has a Hamiltonian cycle. Thus the reduction holds.

Conclusion

The problem is in NP and NP-hard, hence NP-complete.